Pediatric Bio-Responsive Prosthetic Socket with Integrated Pressure Sensing

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Clinical Context

- **66%** of users **experience socket discomfort** [1].
- Limb volume changes affect fit throughout the day.
- Poor fit reduces comfort and usability.

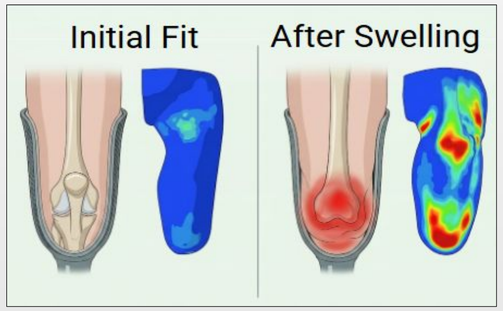

Figure 1. *Pressure distribution comparison showing* ***increased tissue stress*** *after limb swelling.*

Our Approach

We developed a **bio-responsive prosthetic socket** that **automatically adjusts fit** in **response to pressure changes**.

The system combines **embedded sensing** with **mechanical actuation** to maintain consistent comfort.

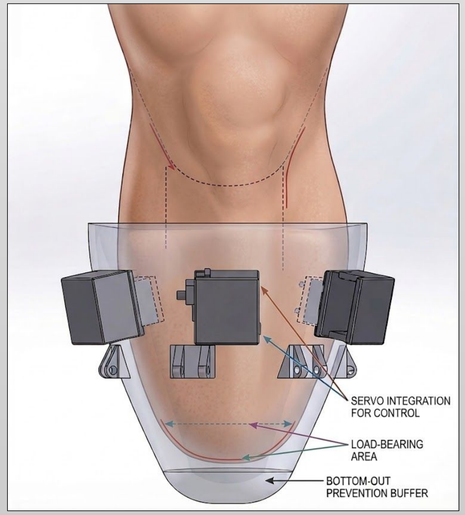

Figure 2. Anatomical Integration. *Interaction between the* ***residual limb and the adaptive socket****, highlighting sensor-driven servo control and stabilized load-bearing zones for patient comfort.*

System Overview

1. **Sensing:** Embedded pressure sensors detect limb swelling.
2. **Control:** A custom PCB processes the data in real-time.
3. **Response:** Mini-servos move internal panels to adjust the fit.
4. **Comfort:** The flexible socket maintains a secure, pain-free grip.

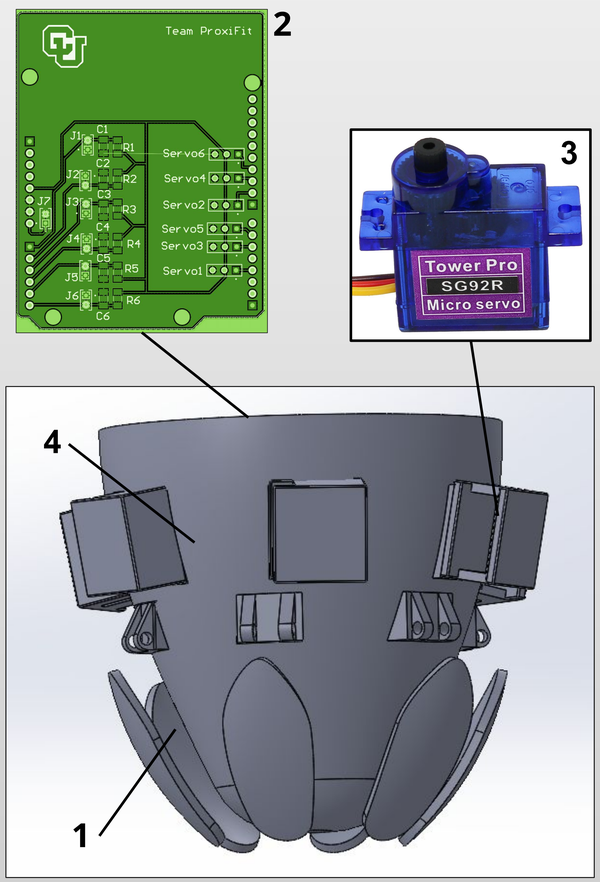

Figure 3. ProxiFit System Prototype. *CAD model illustrating the integrated hardware: (1) Petals with FSRs, (2) custom PCB control unit, (3) mechanical servo actuators, and (4) flexible TPU socket core.*

Results

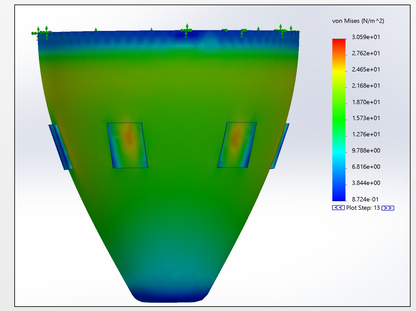

Figure 4. Structural Validation. *FEA simulation confirms the socket withstands required forces while successfully targeting high-pressure zones for relief.*

Designed to reduce peak pressure by up to ~40%

- Successfully **detects localized pressure** changes
- Demonstrates **controlled** mechanical adjustment
- Meets torque and structural requirements

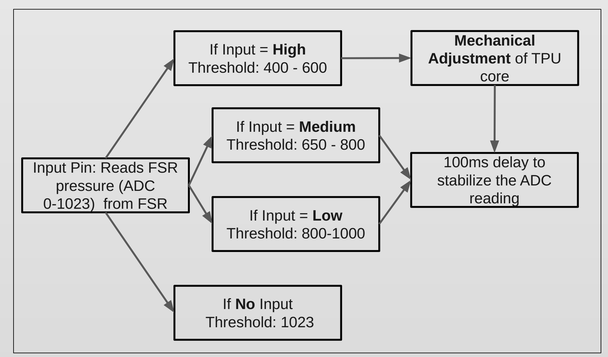

Figure 5. Adaptive Control Logic. *Categorizes real-time pressure data into three levels. When 'High' pressure is detected, the software* ***initiates a mechanical adjustment*** *to restore comfort, maintaining a stable fit through feedback loops.*

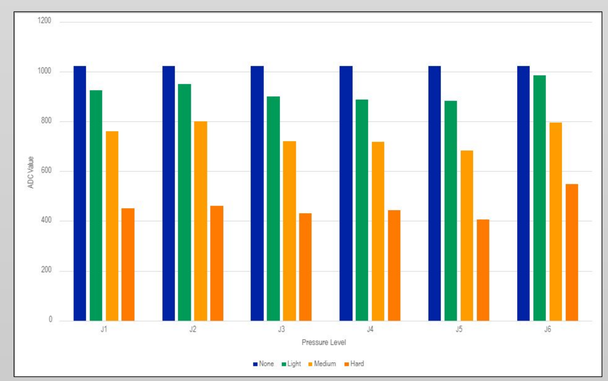

Figure 6. Sensor Calibration & Consistency. *Data used to define the three different thresholds for consistent, automated response [2].*

Recommendations

- **Clinical Testing:** Evaluate pediatric comfort and skin integrity during high-impact activities.
- **Material Optimization:** Refine TPU socket to balance limb stability and weight.
- **Thermal Regulation:** Integrate cooling channels to improve long-term wearability.
- **Modular Scaling:** Adapt the socket geometry to cater a larger population.
- **Aesthetic:** Integrating another configuration for the motors to not be as vulnerable to the environment.

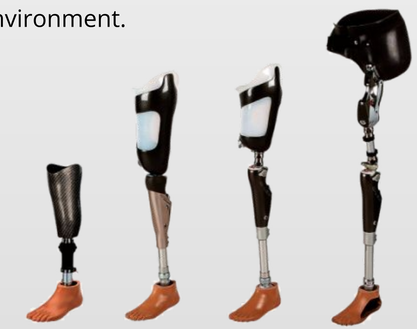

Figure 7. Future Scalability. *The system can be adapted to various prosthetic geometries, ensuring the smart-fit technology is accessible across different amputation levels and patient populations.*

Acknowledgements

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References

[1] "Socket Discomfort After Below-Knee Amputation," *Limber Prosthetics*, May 15, 2025. https://limberprosthetics.com/socket-discomfort-after-below-knee-amputation/ (accessed Mar. 10, 2026).

[2] Adafruit Learning System, "Using an FSR," *Force Sensitive Resistor (FSR)*, Jul. 29, 2012, updated Mar. 8, 2024.